

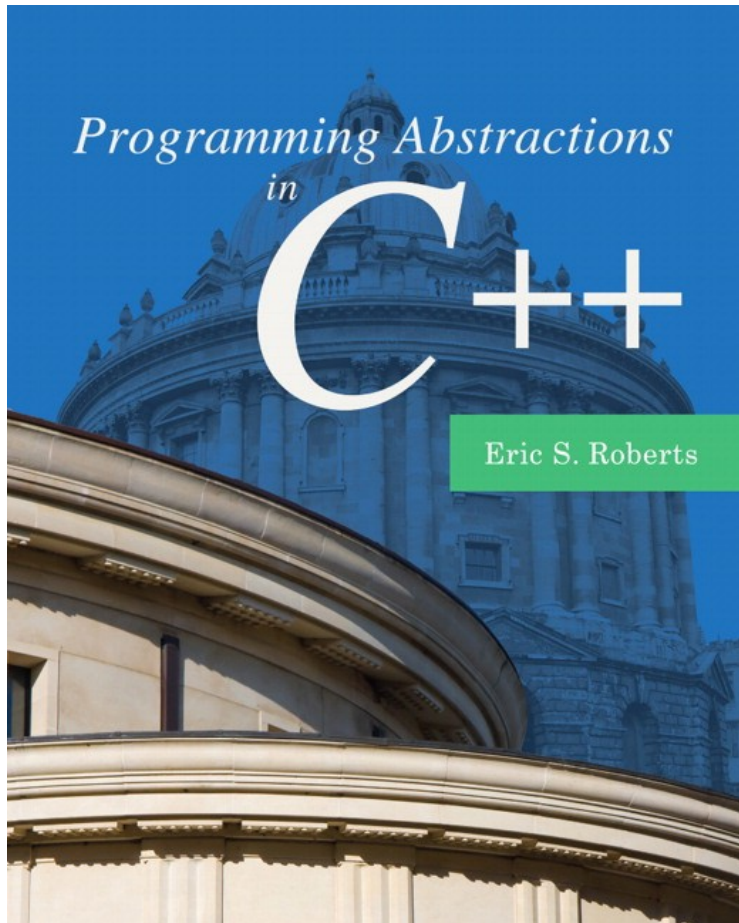
Welcome to CS101B!

Prerequisites

CS101A

(or equivalent)

Our Textbook



- Our course textbook is ***Programming Abstractions in C++*** by the legendary Eric Roberts.
 - There's a **draft version** available online.
- We've assigned readings for each lecture. You can either do them before or after the lectures – your choice.
- It's important to complete the readings in addition to attending lecture; there's a lot of really good info in there.

What's Next in Computer Science?

Goals for this Course

- ***Learn how to model and solve complex problems with computers.***
- To that end:
 - Explore common abstractions for representing problems.
 - Harness recursion and understand how to think about problems recursively.
 - Quantitatively analyze different approaches for solving problems.

Goals for this Course

Learn how to model and solve complex problems with computers.

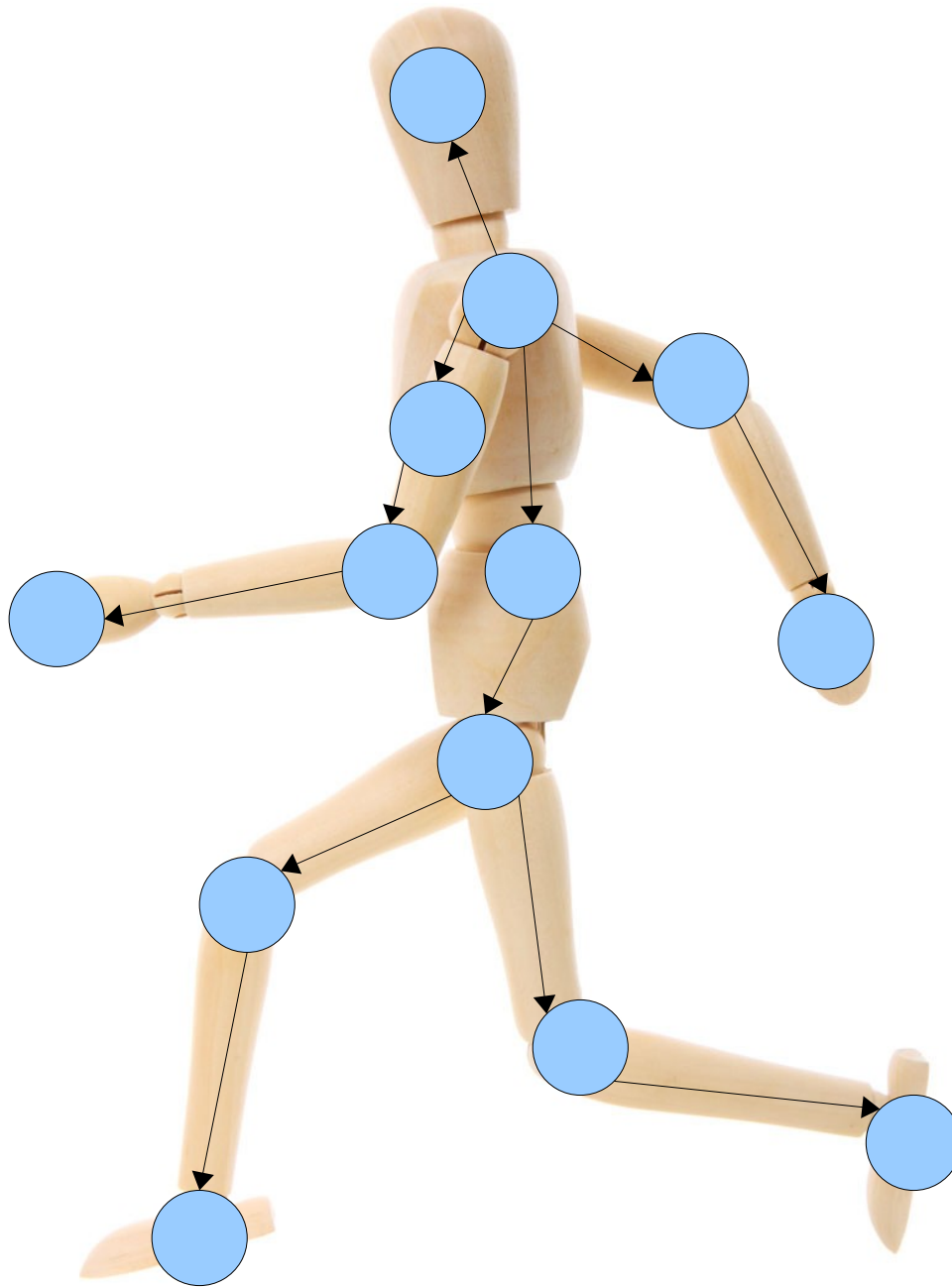
To that end:

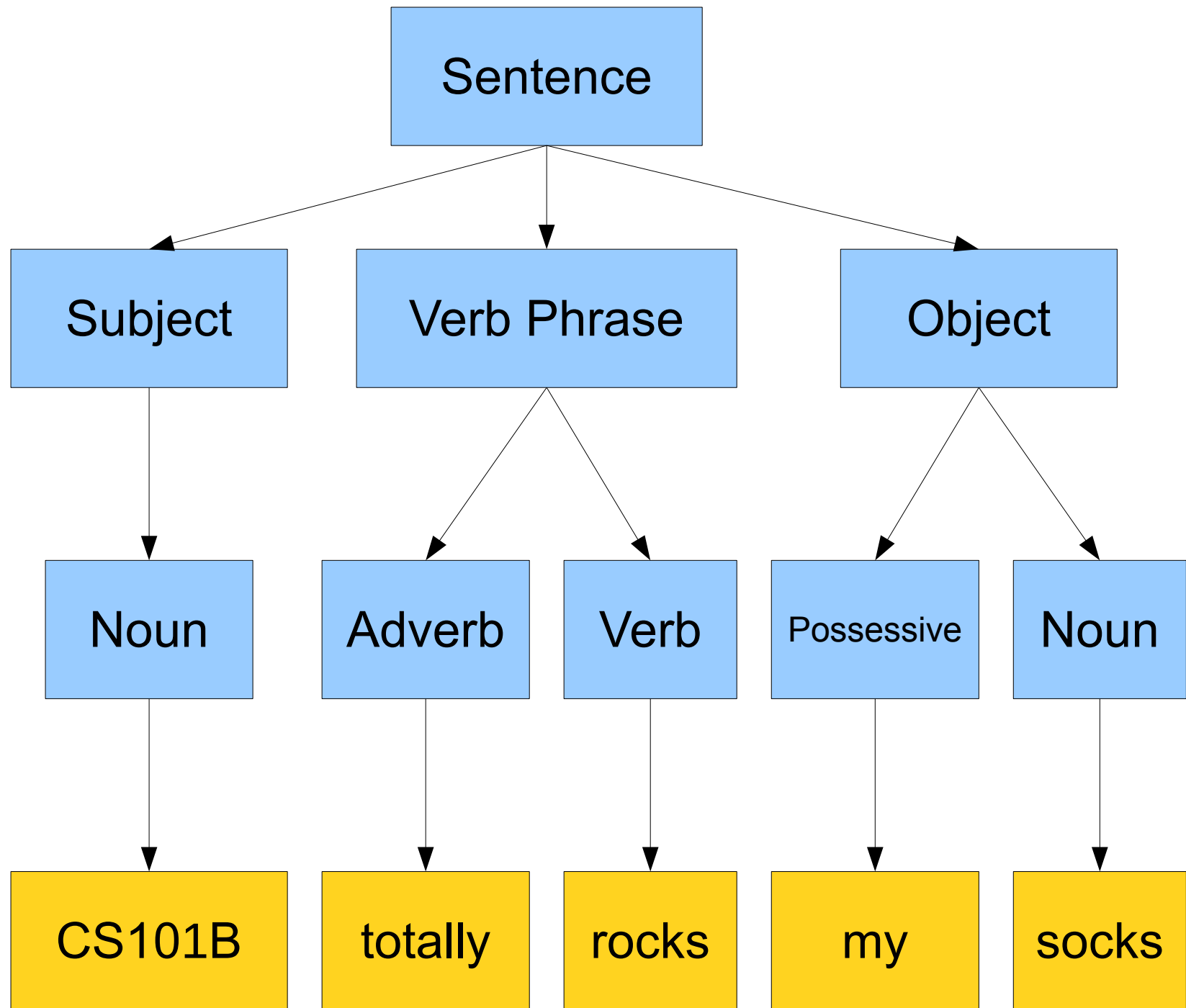
- Explore common abstractions for representing problems.

Harness recursion and understand how to think about problems recursively.

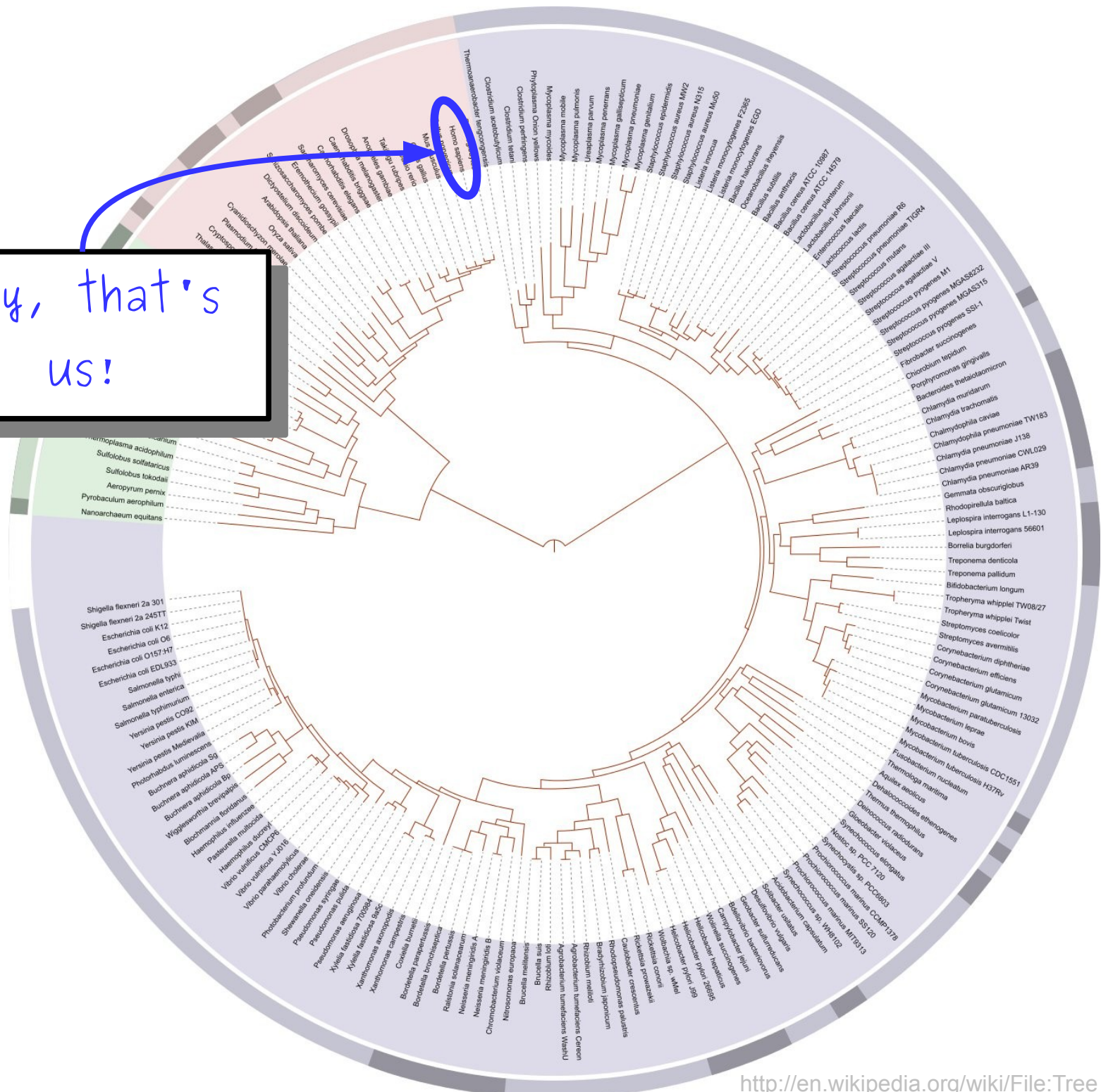
Quantitatively analyze different approaches for solving problems.

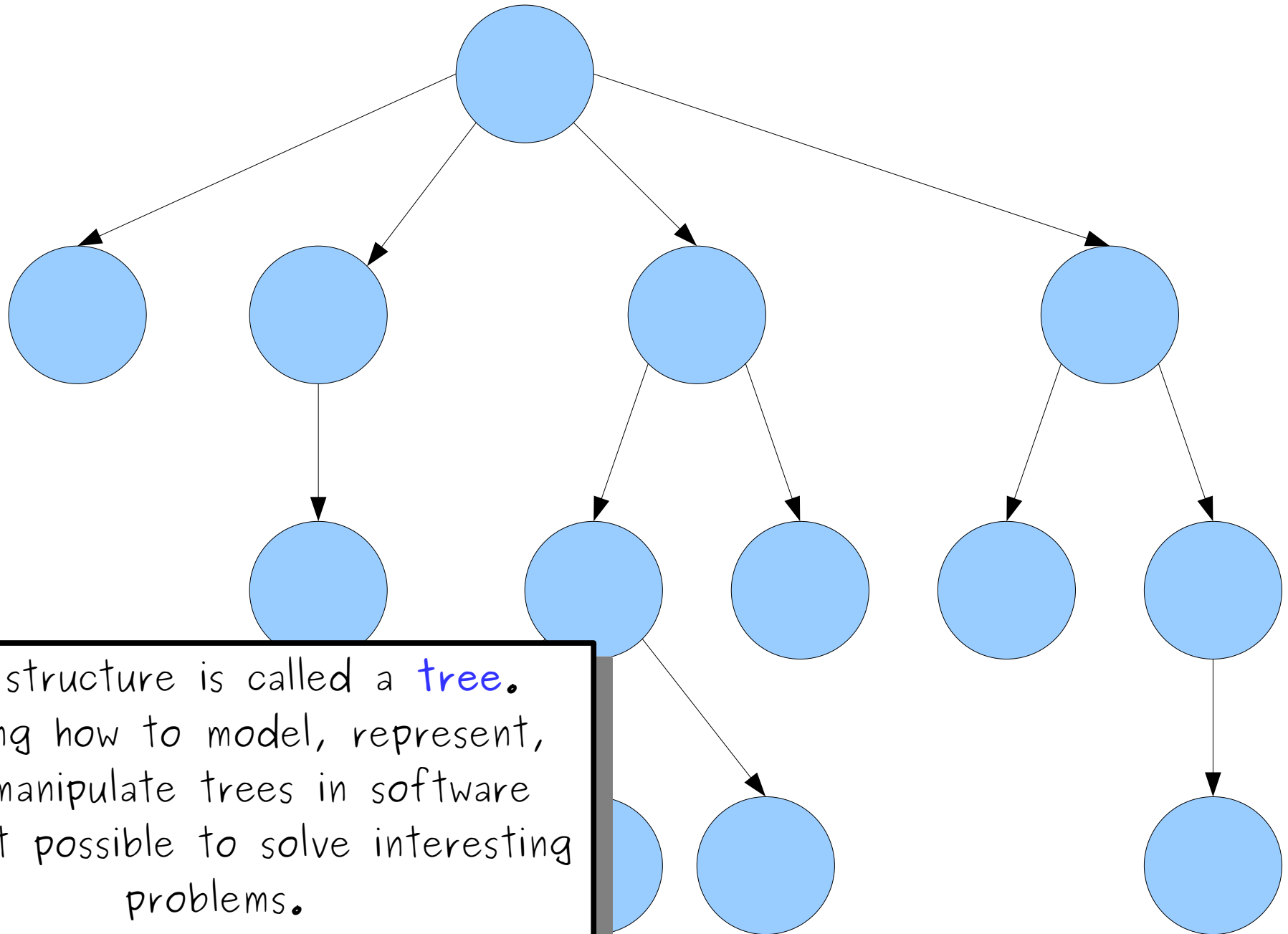






Hey, that's
us!





This structure is called a **tree**.
Knowing how to model, represent,
and manipulate trees in software
makes it possible to solve interesting
problems.

Building a vocabulary of ***abstractions*** makes it possible to represent and solve a wider class of problems.

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 - Quantitatively analyze different approaches for solving problems.

Goals for this Course

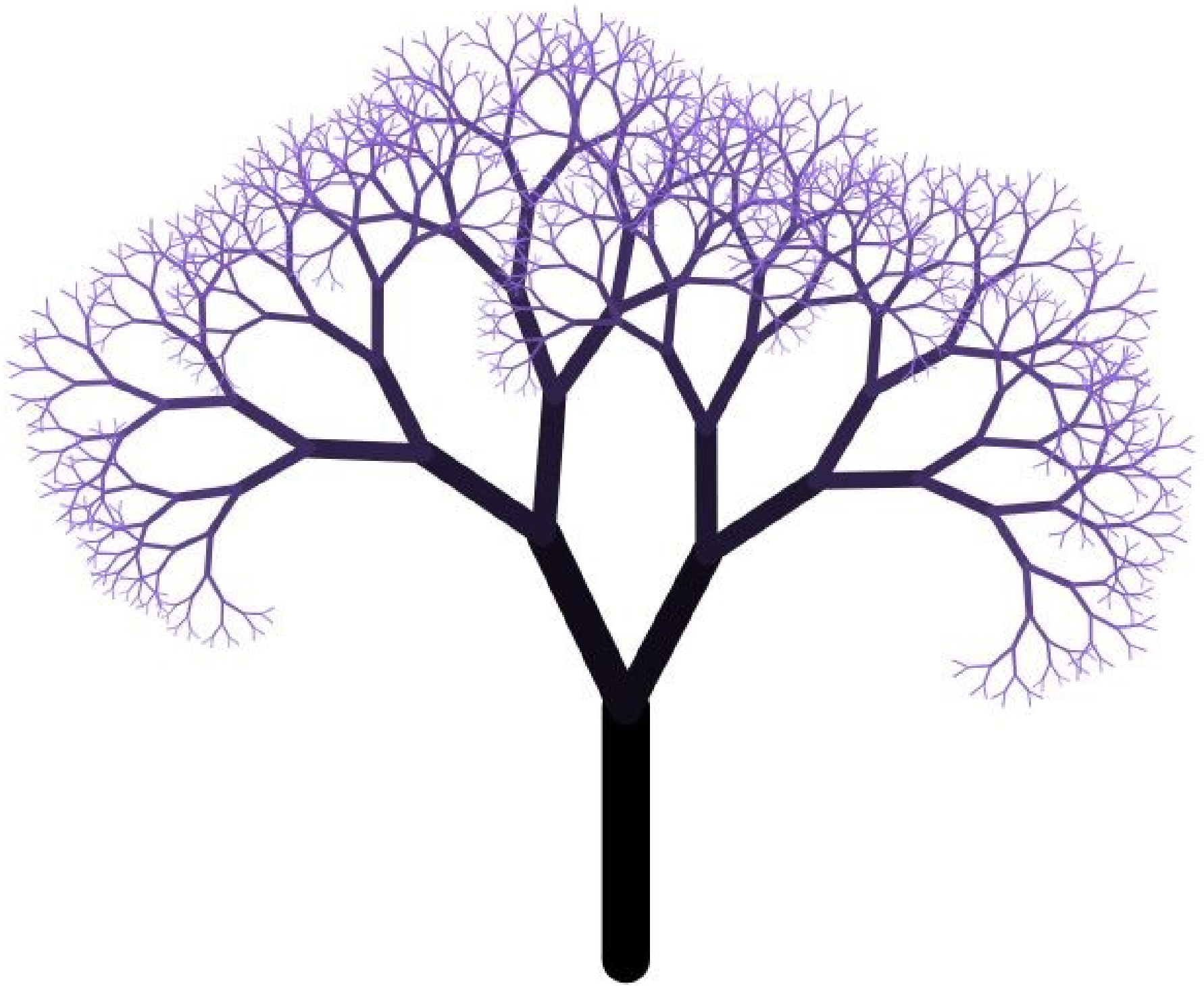
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To that end:

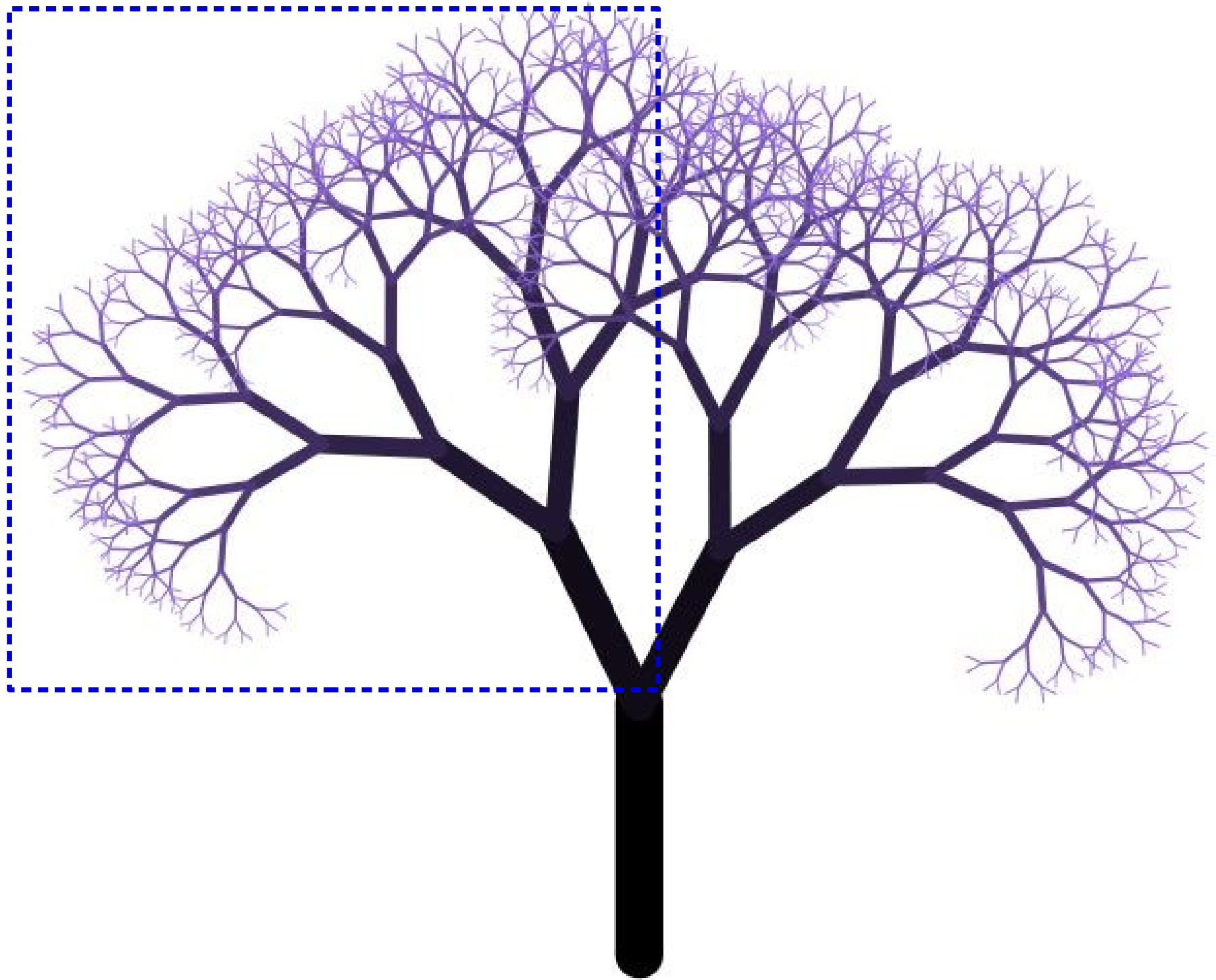
Explore common abstractions for representing problems.

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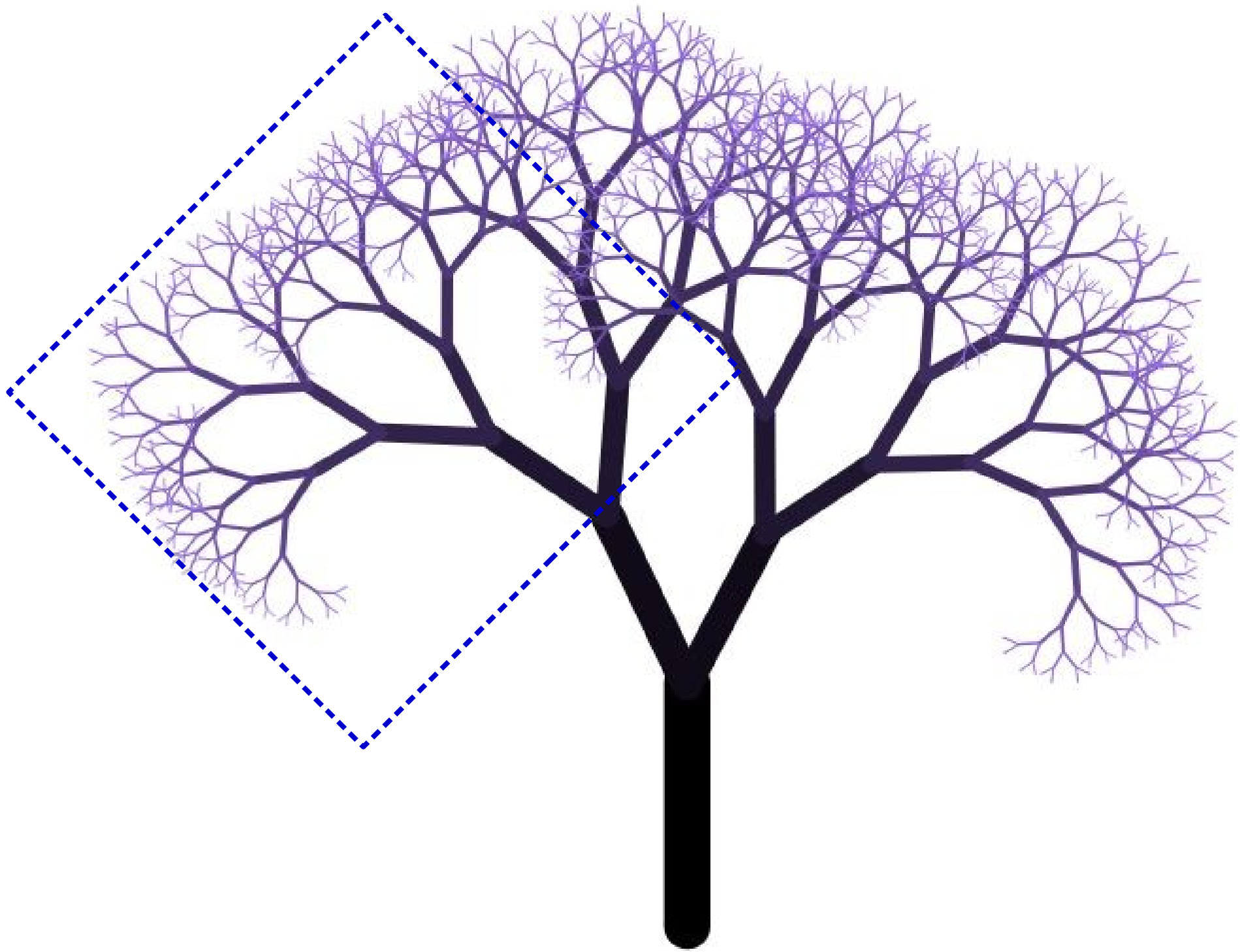
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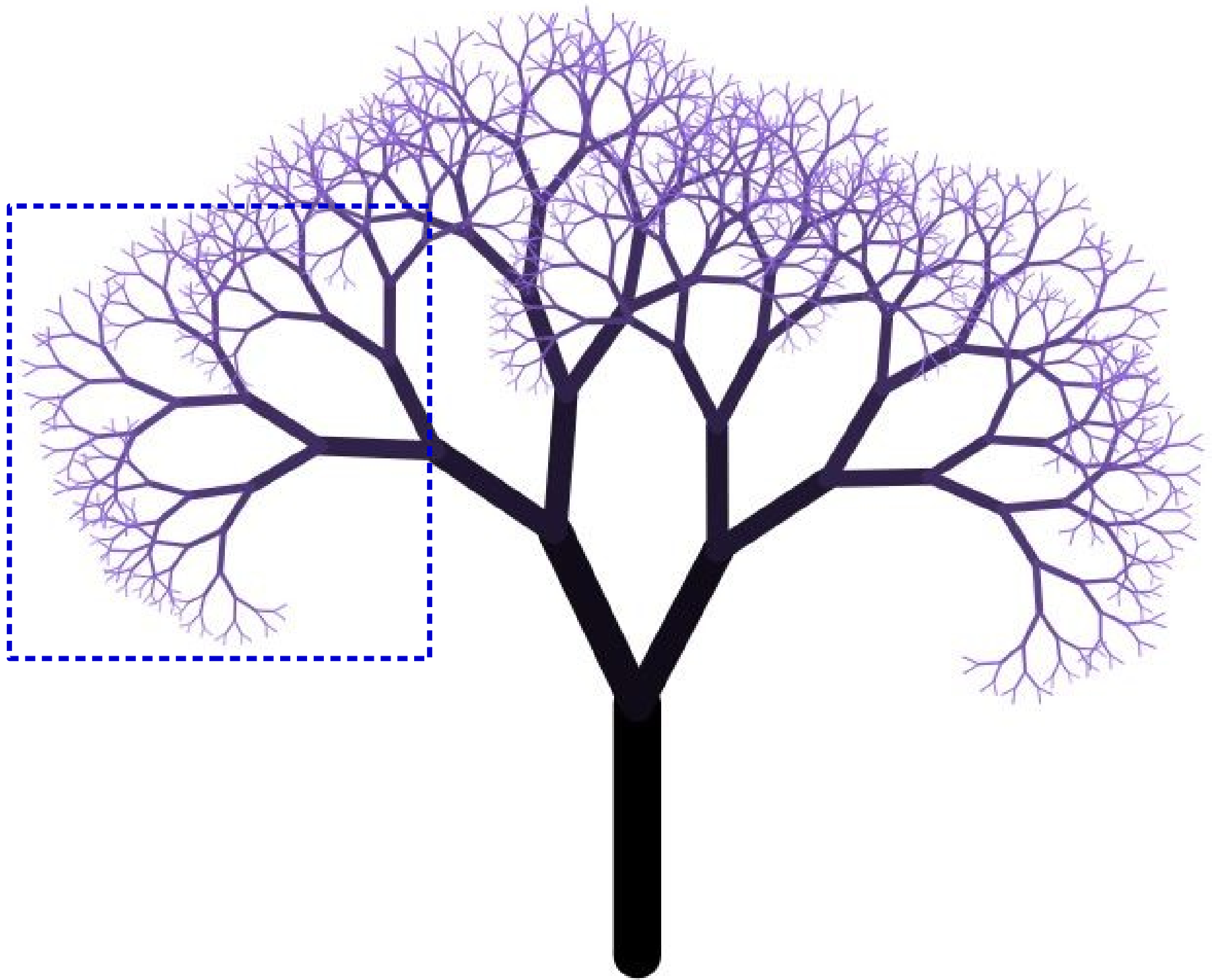
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<http://www.marketoracle.co.uk/images/2010/Oct/fractal-tree2.jpg>

A ***recursive solution*** is a solution that is defined in terms of itself.

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- Quantitatively analyze different approaches for solving problems.

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There are many ways to solve the same problem. How do we ***quantitatively*** talk about how they compare?

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Transitioning to C++

Transitioning to C++

- Assuming that the majority of you are either coming out of CS101A in Python.
- In this course, we'll use the C++ programming language.
- Learning a second programming language is ***substantially*** easier than learning a first.
 - You already know how to solve problems; you just need to adjust the syntax you use.
- While the languages are superficially different, they have much in common.

Our First C++ Program

Prime Numbers

- A positive integer n is called a ***prime number*** if it's greater than one and its only positive divisors are 1 and n .
- For example:
 - 15 isn't prime ($15 = 3 \times 5$).
 - 17 is prime.
 - 19 is prime.
 - 21 isn't prime ($21 = 3 \times 7$).
- What's the 5,000th prime number?

```
def isPrime(n):  
    """Returns whether the input number is prime; assumes n >= 2."""  
  
    # Try dividing by all numbers from 2 to n - 1, inclusive.  
    for divisor in range(2, n):  
        if n % divisor == 0:  
            return False  
  
    return True
```

```
found = 0    # How many prime numbers we've found  
number = 1   # Next number to test  
  
# Keep trying numbers until we've found 5000 primes.  
while found < 5000:  
    number += 1  
    if isPrime(number):  
        found += 1  
  
# Last number tried is the 5000th prime.  
print(number)
```

```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n - 1, inclusive. */
    for (int divisor = 2; divisor < n; divisor++) {
        if (n % divisor == 0) {
            return false;
        }
    }
    return true;
}

int main() {
    int found = 0; // How many prime numbers we've found
    int number = 1; // Next number to test

    /* Keep trying numbers until we've found 5000 primes. */
    while (found < 5000) {
        number++;
        if (isPrime(number)) {
            found++;
        }
    }

    /* Last number tried is the 5000th prime. */
    cout << number << endl;
    return 0;
}

```

```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n - 1, inclusive. */
    for (int divisor = 2; divisor < n; divisor++) {
        if (n % divisor == 0) {
            return false;
        }
    }
    return true;
}

int main() {
    int found = 0; // How many prime numbers we've found
    int number = 1; // Next number to test

    /* Keep trying numbers until we've found 5000 primes. */
    while (found < 5000) {
        number++;

        if (isPrime(number)) {
            found++;
        }
    }

    /* Last number tried is the 5000th prime. */
    cout << number << endl;
    return 0;
}

```

In Python, indentation alone determines nesting.

In C++, indentation is nice, but **curly braces** alone determine nesting.

```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n - 1, inclusive. */
    for (int divisor = 2; divisor < n; divisor++) {
        if (n % divisor == 0) {
            return false;
        }
    }
    return true;
}

```

Python uses **True** and **False**;
C++ uses **true** and **false**.

```

int main() {
    int found = 0; // How many prime numbers we've found
    int number = 1; // Next number to test

    /* Keep trying numbers until we've found 5000 primes. */
    while (found < 5000) {
        number++;

        if (isPrime(number)) {
            found++;
        }
    }

    /* Last number tried is the 5000th prime. */
    cout << number << endl;
    return 0;
}

```



```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n-1. */
    for (int divisor = 2; divisor < n; divisor++)
        if (n % divisor == 0) {
            return false;
        }
    return true;
}

```

In Python, newlines mark the end of statements.

In C++, individual statements must have a semicolon (;) after them.

```

int main() {
    int found = 0; // How many primes found
    int number = 1; // Number to test

    /* Keep trying numbers until we find 5000 primes. */
    while (found < 5000)
        number++;

    if (isPrime(number)) {
        found++;
    }

    /* Last number tried is the 5000th prime. */
    cout << number << endl;
    return 0;
}

```

return false;

```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n - 1, inclusive. */
    for (int divisor = 2; divisor < n; divisor++) {
        if (n % divisor == 0) {
            return false;
        }
    }
    return true;
}

int main() {
    int found = 0; // How many prime numbers we've found
    int number = 1; // Next number to test

    /* Keep trying numbers until we've found 5000 primes. */
    while (found < 5000) {
        number++;
        if (isPrime(number)) {
            found++;
        }
    }

    /* Last number tried is the next prime. */
    cout << number << endl;
    return 0;
}

```

In Python, you print output by using `print()`.

In C++, you use the ***stream insertion operator*** (`<<`) to push data to the console. (Pushing `endl` prints a newline.)

```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n-1 */
    for (int divisor = 2; divisor < n; divisor++) {
        if (n % divisor == 0) {
            return false;
        }
    }
    return true;
}

int main() {
    int found = 0; // How many prime numbers we've found
    int number = 1; // Next number to test

    /* Keep trying numbers until we've found 5000 primes. */
    while (found < 5000) {
        number++;

        if (isPrime(number)) {
            found++;
        }
    }

    /* Last number tried is the 5000th prime. */
    cout << number << endl;
    return 0;
}

```

In Python, you can optionally put parentheses around conditions in if statements and while loops.

In C++, these are mandatory.

```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n - 1, inclusive. */
    for (int divisor = 2; divisor < n; divisor++) {
        if (n % divisor == 0)
            return false;
    }
    return true;
}

```

Python and C++ each have **for** loops, but the syntax is different. (Check the textbook for more details about how this works!)

```

int main() {
    int found = 0; // How many prime numbers we've found
    int number = 1; // Next number to test

    /* Keep trying numbers until we've found 5000 primes. */
    while (found < 5000) {
        number++;

        if (isPrime(number)) {
            found++;
        }
    }

    /* Last number tried is the 5000th prime. */
    cout << number << endl;
    return 0;
}

```

```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n - 1, inclusive. */
    for (int divisor = 2; divisor < n; divisor++) {
        if (n % divisor == 0) {
            return false;
        }
    }
    return true;
}

int main() {
    int found = 0; // How many primes found
    int number = 1; // Next number to test

    /* Keep trying numbers until we've found 5000 primes. */
    while (found < 5000) {
        number++;
        if (isPrime(number)) {
            found++;
        }
    }

    /* Last number tried is the 5000th prime. */
    cout << number << endl;
    return 0;
}

```

C++ has an operator ++ that means “change this variable’s value by adding one to it.” Python doesn’t have this.

```
#include <iostream>
using namespace std;
```

```
/* Returns whether the input number is prime; assumes n >= 2. */
```

```
bool isPrime(int n) {
```

```
/* Try dividing by all numbers from 2 to n - 1, inclusive. */
```

```
for (int divisor = 2; divisor < n; divisor++) {
```

```
    if (n % divisor == 0)
```

```
        return false;
```

```
    }
```

```
}
```

```
return true;
```

```
}
```

```
int main() {
```

```
    int found = 0; // How many prime numbers we've found
```

```
    int number = 1; // Next number to test
```

```
/* Keep trying numbers until we've found 5000 primes. */
```

```
while (found < 5000) {
```

```
    number++;
```

```
    if (isPrime(number)) {
```

```
        found++;
```

```
    }
```

```
}
```

```
/* Last number tried is the 5000th prime. */
```

```
cout << number << endl;
```

```
return 0;
```

```
}
```

In Python, comments start with # and continue to the end of the line.

In C++, there are two styles of comments. Comments starting with /* continue until */. Comments starting with // continue to the end of the line.

```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n - 1, inclusive. */
    for (int divisor = 2; divisor < n;
        if (n % divisor == 0) {
            return false;
        }
    }
    return true;
}

int main() {
    int found = 0; // How many prime numbers we've found
    int number = 1; // Next number to test

    /* Keep trying numbers until we've found 5000 primes. */
    while (found < 5000) {
        number++;

        if (isPrime(number)) {
            found++;
        }
    }

    /* Last number tried is the 5000th prime. */
    cout << number << endl;
    return 0;
}

```

In Python, comments on functions are customarily written after the first line of the function.

In C++, comments on functions are customarily written *before* the first line of the function.

```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n - 1 inclusive. */
    for (int divisor = 2; divisor < n; divisor++)
        if (n % divisor == 0) {
            return false;
        }
    return true;
}

int main() {
    int found = 0; // How many prime numbers we've found
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    /* Keep trying numbers until we've found 5000 primes. */
    while (found < 5000) {
        number++;

        if (isPrime(number)) {
            found++;
        }
    }

    /* Last number tried is the 5000th prime. */
    cout << number << endl;
    return 0;
}

```

In Python, each object has a type, but it isn't stated explicitly.

In C++, you *must* give a type to each variable. (The **int** type represents an integer, and **bool** represents a Boolean true or false.)


```

#include <iostream>
using namespace std;

/* Returns whether the input number is prime; assumes n >= 2. */
bool isPrime(int n) {
    /* Try dividing by all numbers from 2 to n-1. */
    for (int divisor = 2; divisor < n; divisor++)
        if (n % divisor == 0) {
            return false;
        }
    return true;
}

```

In Python, statements can be either in a function or at the top level of the program.

In C++, most statements must be inside of a function.

```

int main() {
    int found = 0; // How many prime numbers we've found
    int number = 1; // Next number to test

    /* Keep trying numbers until we've found 5000 primes. */
    while (found < 5000) {
        number++;

        if (isPrime(number)) {
            found++;
        }
    }

    /* Last number tried is the 5000th prime. */
    cout << number << endl;
    return 0;
}

```

Why do we have both C++ and Python?

C++ and Python

- Python is a *great* language for data processing and writing quick scripts across all disciplines.
 - It's pretty quick to make changes to Python programs and then run them to see what's different.
 - Python programs, generally, run more slowly than C++ programs.
- C++ is a *great* language for writing high-performance code that takes advantage of underlying hardware.
 - Compiling C++ code introduces some delays between changing the code and running the code.
 - C++ programs, generally, run much faster than Python programs.
- Knowing both languages helps you use the right tool for the right job.

Your Action Items

- ***Read Chapter 1 of the textbook.***
 - Use this as an opportunity to get comfortable with the basics of C++ programming and to read more examples of C++ code.
- ***Start Assignment 0.***
 - The assignment and its starter files are up on the course website.
 - No programming involved, but you'll need to get your development environment set up.
 - There's a bunch of documentation up on the course website. Please feel free to reach out to us if there's anything we can do to help out!

Next Time

- ***Welcome to C++!***
 - Defining functions.
 - Basic arithmetic.
 - Writing loops.